

10

Department of Education
National Capital Region
SCHOOLS DIVISION OFFICE
MARIKINA CITY

MATHEMATICS

Quarter 3: Module 1

Permutation



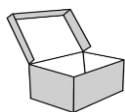
Writer: Christlie Rose R. Nicolas

Cover Illustrator: Joel J. Estudillo



City of Good Character
DISCIPLINE • GOOD TASTE • EXCELLENCE

Government Property
NOT FOR SALE



What I Need to Know

Hello, Grade 10 learners! In this module, you will learn how to:

- a) Illustrate the permutation of objects (M10SP-IIIa-1)**
- b) Solve problems involving permutations (M10SP-IIIb-1)**

You can say that you have understood the lesson in this module if you can already:

1. illustrate permutation of objects in various forms: linear, circular, and distinguishable;
2. solve problems involving permutations.



What I Know

Directions: Choose the letter that corresponds to the correct answer.

1. Which of the following is the arrangement of n objects in a specified order?
A. event
B. combination
C. permutation
D. sample space
2. If there are 7 people and only 5 chairs are available, which of the following is the correct illustration of how many ways can they be seated?
A. 7×5
B. 5×7
C. $7 \times 6 \times 5 \times 4 \times 3$
D. $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$
3. How many ways can five different bicycles be lined up in a parking space?
A. 120
B. 60
C. 25
D. 5
4. Evaluate: $P(7, 5)$
A. 21
B. 35
C. 250
D. 2520
5. A bracelet has 6 different charms. How many arrangements of the charms are possible?
A. 60
B. 120
C. 720
D. 5040



6. In how many ways can Chris and her four friends be seated in a round table?

A. 3!	C. 5!
B. 4!	D. 6!
7. If friends of 5 attend a party and two of them must sit together in a circular table, what must be the value of n ?

A. 5	C. 3
B. 4	D. 2
8. How many distinguishable permutations can be made from the letters of the word LOVE?

A. 12	C. 20
B. 18	D. 24
9. You have 3 identical bags and 4 identical socks to be given to your 7 friends. In how many ways can you distribute those 7 gifts to your 7 friends?

A. 35	C. 56
B. 49	D. 89
10. Which of the following illustrates permutation?
 - A. Choosing a song to play
 - B. Assigning LRN among the students
 - C. Choosing a guest to attend in a party
 - D. Selecting a hero in playing Mobile Legends



Lesson 1. Linear Permutation



What's In

In your Grade 8 class, you have learned that the total number of ways different events can occur is known as **Fundamental Counting Principle (FCP)**. This concept is important and helpful for us to learn our lesson in this module.

Let us try answering this situation:

Chris has 8 different books, but his bookshelf can only occupy 5 books. In how many ways can he arrange those books in the shelf?

You may use listing method or draw an illustration to help you answering the problem.

Likewise, let us also recall an important symbol which will be used throughout this module.

n!

Some students say that this symbol pertains to a number that is always angry because of the exclamation point beside it. But this is called as **factorial notation** or read as "**n-factorial**".

Factorial notation is a product of all positive integers less than or equal to n . It has a special case where $0! = 1$.

Examples:

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$3! + 4! = 6 + 24 = 30$$

$$(2!) (3!) = (2 \times 1) (3 \times 2 \times 1) = (2) (6) = 12$$



What's New

City of Good Character

DISCIPLINE • GOOD TASTE • EXCELLENCE

Before discussing deeper our main lesson in this module, let us activate our minds through this activity called *anagrams*. It is a type of word play wherein the rearrangement of letters will produce a new word.

Find anagrams of the following words. (Hint: Body parts)

sink	_____
slip	_____
earth	_____
fringe	_____
impart	_____

Guide Questions:

- 1) Based on the activity, how did you form new words?
- 2) Is order important in forming new words?
- 3) Using this activity, what is the concept of linear permutation?



What is It

Unknowingly, as we arrange letters, situations, or things, we are already applying a concept in Mathematics. If **order** matters in an arrangement, this is called as **permutation**.

Permutation has three kinds of arrangements, namely: linear, circular, and distinguishable permutation. This will be discussed further on each lesson.

Definition: **LINEAR PERMUTATION**

It refers to the number of ordered arrangements in a *line*.

Symbol: **nPr** or **$P(n, r)$**

Read as: “the number of permutations of **n** objects taken **r** at a time”

Any objects or persons involved that can be arranged forming a line is an illustration of a linear permutation.

For example:



- 1) Arrangement of books in a shelf
- 2) Arrangement of names in class record
- 3) Arrangement of students from smallest to highest in height

In getting the total number of ways can be arranged in a linear permutation, use this formula:

LINEAR PERMUTATION FORMULA:

where: n – total number of objects
 r – objects taken for arrangement
 $n \geq r$

Example 1: Marikina celebrates Sapatos Festival. In a certain booth, there are 10 new shoe designs to be displayed. In how many ways can these shoes be arranged if there are only 7 spaces available in a row?

Explanation: The arrangement of shoes in a rack is in linear form.

Solution: $n = 10$; $r = 7$

5

$${}_nP_r = \frac{n!}{(n-r)!} \quad \text{Formula in Linear Permutation}$$

$${}_{10}P_7 = \frac{10!}{(10-7)!} \quad \text{Substitute the values of } n \text{ and } r$$

$${}_{10}P_7 = \frac{10!}{(3)!} \quad \text{Simplify } (10 - 7)!$$

$${}_{10}P_7 = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3!}{(3)!} \quad \text{Expand } 10!$$

$${}_{10}P_7 = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \quad \text{Cancel } 3! \text{ in numerator and denominator}$$

$${}_{10}P_7 = 604\,800 \quad \text{Simplify}$$

Answer: **There are 604 800 ways in arranging 10 new shoe designs if there are only 7 spaces.**

Example 2: How many ways can a chairperson and a co-chairperson be chosen from 15 active members in Math Club?

Explanation: The problem illustrates a linear permutation because there is an indicated order, specifically a position to consider.



Solution: $n = 15$; $r = 2$

$${}_nP_r = \frac{n!}{(n-r)!}$$

Formula in Linear Permutation

$${}_{15}P_2 = \frac{15!}{(15-2)!}$$

Substitute the values of n and r

$${}_{15}P_2 = \frac{15!}{(13)!}$$

Simplify $(15 - 2)!$

$${}_{15}P_2 = \frac{15 \times 14 \times 13!}{(13)!}$$

Expand $15!$

$${}_{15}P_2 = 15 \times 14$$

Cancel $13!$ in numerator and denominator

$${}_{15}P_2 = 210$$

Simplify

Answer: **There are 210 ways.**

Example 3: How many ways can 5 men and 4 women be arranged in a row alternately?

Explanation: The term “row” means arrangement in a linear form.

Illustration:



Note: Since we have two categories, man and woman, there will also be two parts of the solution.

Solution: For Men: $n = 5$; $r = 5$

For Women: $n = 4$; $r = 4$

$${}_nP_r = \frac{n!}{(n-r)!}$$

$${}_nP_r = \frac{n!}{(n-r)!}$$

$${}_5P_5 = \frac{5!}{(5-5)!}$$

$${}_4P_4 = \frac{4!}{(4-4)!}$$

$${}_5P_5 = \frac{5!}{(0)!}$$

$${}_4P_4 = \frac{4!}{(0)!}$$

$${}_5P_5 = \frac{5!}{1}$$

$${}_4P_4 = \frac{4!}{1}$$

$${}_5P_5 = 5!$$

$${}_4P_4 = 4!$$

$${}_5P_5 = 120$$

$${}_4P_4 = 24$$

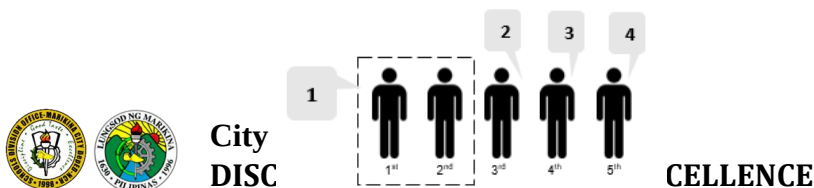
$${}_5P_5 \times {}_4P_4 = 120 \times 24 = 2880$$

Answer: **There are 2,880 ways.**

Example 4: In how many ways can 5 contestants be lined up in a stage if two of them insisted to stand together?

Explanation: The term “lined up” indicates a linear form of arrangement.

Illustration:



Note: Supposedly the value of “n” is 5 because of ‘5 contestants’. However, there are restrictions that ‘*two of them insisted to stand together*’. So, the two contestants who insisted to be together will be counted as one. This indicates that $n = 4$.

Solution: $4P_4$ = means for four indicated positions

$2P_2$ = means the two people can still exchange their places

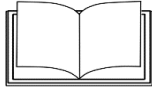
$$4P_4 \times 2P_2 = 24 \times 2 = 48$$

Permutation can easily be computed using scientific calculator. Just look for the following symbols:

Most of the times it is written in between buttons.

Answer: ***There are 48 ways.***





What's More

A. Write **P** if the situation illustrates permutation. If not, write **NP**.

1. Entering a password in e-learning platform
2. Answering a multiple-choice test
3. Drawing a name in the raffle
4. Arrangement of shoes in the rack
5. Checking the class attendance

B. Evaluate the following permutations.

1. ${}_6P_5 = \underline{\hspace{2cm}}$
2. ${}_7P_7 = \underline{\hspace{2cm}}$
3. ${}_3P_1 = \underline{\hspace{2cm}}$
4. ${}_9P_2 = \underline{\hspace{2cm}}$
5. ${}_2P_5 = \underline{\hspace{2cm}}$

C. Answer the following problems with complete solution.

1. In how many ways may 4 people be seated in a row of 10 chairs?
2. In how many ways can a point guard, shooting guard, small forward, power forward, and center can be selected from a basketball team having ten players?
3. In how many ways can you arrange four persons in a row for a picture taking if two people insisted to be together?



What I Have Learned

Write "**Fact**" if the statement is correct, otherwise, write "**BLUFF**".

1. Linear permutation refers to the number of ordered arrangements of different objects in a line.
2. The value of "**n**" should be less than or equal to "**r**".
3. The general formula for permutation is ${}_nP_r = \frac{n!}{(n-r)!}$.
4. The symbol "**r**" is the total number of items in the sample.
5. Permutation is a selection of n objects where the order does matter.





What I Can Do

“PERFORMANCE TASK: PRIORITIES”

Goal: Make a to-do list for the day.

Role: You are a responsible student.

Audience: Classmates, teacher, and your parents/guardians.

Situation: You need to do a lot of things for the day other than the things to be done for school. The list shows the options:

- Clean your room.
- Do the activities in module.
- Attend a birthday dinner of your grandparent.
- Play a mobile game.
- Watch Korean drama or movie.
- Read a lesson for advance study.
- Help in your family business.

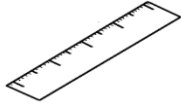
Product: The expected output must be the following:

- 1) List down your top five (5) priorities for the day.
- 2) Explain in 3-5 sentences the reason of your top 5 priorities.
- 3) Answer: In how many ways can you arrange your top 5 priorities for a day? Illustrate your answer using permutation with complete solution.

Standard for Success: Be guided with the scoring rubrics.

CRITERIA	5	3	1	0
Computation	All representation and computations are correct and accurate.	Committed 1 error .	Committed 2-3 errors .	Committed 4 or more errors .
Details and Explanation	Completed the top priorities and explained it logically and clearly.	Completed the top priorities and explained it generally.	Incomplete output.	No output.





Assessment

Directions: Choose the letter that corresponds to the answer.

- What is an arrangement of n objects in a given order?
A. sample space
B. permutation
C. event
D. combination
- All of the following situations illustrate permutation, EXCEPT _____.
A. electing officers in a room
B. arrangement of books in a shelf
C. assigning telephone numbers to subscribers
D. forming a committee from the members of the club
- Evaluate: ${}_4P_3$
A. 3
B. 7
C. 12
D. 24
- In a group of 7 students, the teacher wishes to select 5 students and arrange them in a line. In how many ways can this be done?
A. 24
B. 40
C. 2520
D. 6720
- How many ways can 4 different Math books and 5 different Science books be arranged on a shelf if books of the same subject are together?
A. ${}_4P_4 \times {}_5P_5 \times {}_2P_2$
B. ${}_4P_4 \times {}_5P_5$
C. ${}_9P_9$
D. ${}_5P_4$



Additional Activities

Solve the following problems.

- In how many ways can you arrange the five vowels **a, e, i, o, u** in a line?
- How many ways can seven different cars be lined up if there are five available parking spaces?
- There are 9 different novel books but only 5 books can be arranged in a shelf. How many arrangements can be made?



Lesson 2. Circular Permutation



What's In

Keep in mind the concept of permutation, specifically linear permutation. With this, let us examine how much you remembered.

Identify the following situations if it illustrates permutation or not. Write **P** if it illustrates permutation; otherwise, **NP**.

1. Selecting Christmas ball to decorate
2. Arrangement of shoes in cabinet
3. Class schedule
4. Identifying the top 10 students
5. Selecting fruit in a basket



What's New

Not all things can be arranged in linear form. Observe the following pictures:



Author: Jay Pogi

Sources:

<https://images.app.goo.gl/jRQFrpj5BtUDoD6j8>, <https://images.app.goo.gl/DLuTZRhMNYaFbLgy8>



Reflect on these questions:

- 1) How are the objects arranged in the picture?
- 2) Name at least three objects in your house that were arranged on the same way.





What is It

Besides linear permutation, we also need to learn the number of ways an object to be arranged in a circular way, which is known as *CIRCULAR PERMUTATION*.

Definition: CIRCULAR PERMUTATION

It refers to the number of ordered arrangements in a *circular form*.

Formula:

where n is the total number of objects

Objects arranged in a circular form are hard to distinguish the arrangements unless one of the objects is “fixed”. That is the reason why there is “1” in the formula.

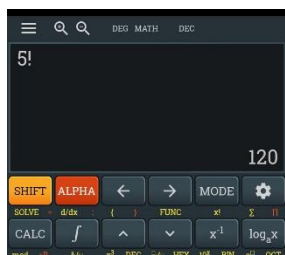
Example 1: If there are six different colored flowers decorated on top of a round cake, in how many ways can these flowers be arranged?

Explanation: Since the cake is round, it gives you the hint that we are dealing with circular permutation. Take note as well with the term “different” because if all objects to be arranged are similar, the number of ways is only one.

Solution: $n = 6$ (six different colored flowers)

$$\begin{aligned}(n - 1)! &= (6 - 1)! && \text{Substitute } n = 6 \\ &= (5)! && \text{Simplify: } 6 - 1 \\ &= 5 \times 4 \times 3 \times 2 \times 1 && \text{Expand } 5! \\ &= 120\end{aligned}$$

Answer: **There are 120 ways.**



You can simply encode $5!$ in your scientific calculator. Look for the symbol of factorial notation as exclamation point (!).

Example:

Press “5” Press “!” Press “=”

If the symbol of “!” is located on top of a button, press “SHIFT”.



Example 2: A bracelet has eight different charms. How many arrangements of charms are possible?

Explanation: A bracelet is obviously in circular form. But take note that it is as simple as the first example. If the problem pertains to a beaded bracelet, the formula on circular permutation must be divided by 2 because a bracelet has two similar phases once worn.

Solution: $n = 8$ (six different colored flowers)

$$\begin{aligned} \frac{(n-1)!}{2} &= \frac{(8-1)!}{2} && \text{Substitute } n = 8 \\ &= \frac{(7)!}{2} && \text{Simplify: } 8 - 1 \\ &= \frac{5040}{2} && \text{Simplify } 7! \\ &= 2520 && \text{Divide 5040 by 2} \end{aligned}$$

Answer: **There are 2520 arrangements.**

Example 3: In how many ways can 10 people be seated at a round table if:

- they can sit anywhere?
- two particular people must sit next to each other?
- two particular people must **not** sit next to each other?

Solutions:

- they can sit anywhere?

Explanation: Since they can sit anywhere, there are no restrictions. Just apply the formula of circular permutation whose $n = 10$, where n is the total number of people seated at a round table.	Solution: $(n - 1)! = (10 - 1)!$ <i>Substitute $n = 10$</i> $= (9)!$ <i>Simplify: $10 - 1$</i> $= 362\,880$ <i>Simplify $9!$</i> Answer: There are 362,880 ways of arranging 10 people seated at a round table.
--	---

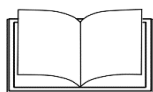


b) two particular people must sit next to each other?

Explanation: There are restrictions on the problem where “two people must sit next to each other”. With this, instead of having $n = 10$, the value of n will become 9 because two people will be counted as one. Then, these two people has their own arrangement as well.	Solution: $(n - 1)! = (9 - 1)! \times 2!$ <i>Substitute $n = 9$ & multiply $2!$ for the arrangement of two people</i> $= (8)! \times 2!$ <i>Simplify: $9 - 1$</i> $= 80\,640$ <i>Simplify $8! \times 2!$</i> Answer: <i>There will be 80,640 ways when 2 out of 10 people insisted to be together in a round table.</i>
---	--

c) two particular people must **not** sit next to each other?

Explanation: This case has also restrictions where “two people must not sit next to each other”. We can apply the concept of the complement of a set. We can subtract the total number of ways where two people must sit next to each other from the times where there are no restrictions.	Solution: $(10 - 1)! - (9 - 1)! \times 2!$ <i>Substitute $n = 10$</i> $= (9)! - 8! \times 2!$ <i>Simplify: $(10 - 1)$ & $(9 - 1)$</i> $= 362\,880 - 80\,640$ <i>Simplify $9!$ & $8! \times 2!$</i> $= 282\,240$ Answer: <i>There are 282,240 ways when two people must not sit next to each other.</i>
--	--



What's More

Solve the following problems.

- How many ways can five different keys be arranged in a key ring?
- A bracelet has five different charms. How many arrangements of charms are possible?
- In how many ways can six different trees be planted in a circular garden?
- In how many ways can 8 people be seated at a round table if:
 - they can sit anywhere?
 - three particular people must sit next to each other?





What I Have Learned

Fill in the blanks with the correct word/phrases.

1. _____ permutation is the numbered of ordered arrangements in circular form.
2. The formula of circular permutation is _____.
3. When encountering a problem about bracelet, the formula of circular permutation must be divided by _____.
4. When two out of 7 people insisted to be together seated in a round table, the value of “n” is _____.



What I Can Do

“PERFORMANCE TASK: PIECE OF HEART”

Goal: Create a beaded bracelet design for your loved one.

Role: You will be a bracelet designer.

Audience: Classmates, teacher and your chosen loved one.

Situation: You decided to make unique gift for your loved one. From six different beads, you need to create your own design.

Product: The expected output must be the following:

- 1) Draw the bracelet with six different design of beads.
- 2) Answer: How many arrangements of beads are possible in a bracelet if there are 6 different designs of beads? Illustrate your answer using permutation with complete solution.

Standard for Success: Be guided with the scoring rubrics.

CRITERIA	5	3	1	0
Computation x 2	All representation and computations are correct and accurate.	Committed 1 error .	Committed 2-3 errors .	Committed 4 or more errors .
Presentation x 2	Presentation of output is unique, creative and neatly done.	Presentation is creative and neatly done.	Incomplete output.	No output.





Assessment

Directions. Choose the letter that corresponds to the correct answer.

- Which of the following pertains to the ordered arrangement of objects in circular form?
A. circular permutation
B. distinguishable permutation
C. linear permutation
D. none of the above
- Which of the following is the correct formula when solving circular permutation?
A. $n!$
B. $n - 1!$
C. $(n - 1)!$
D. $n - r!$
- In how many ways can Chris and her 7 friends be seated in a circular table?
A. $6!$
B. $7!$
C. $8!$
D. $9!$
- How many arrangements are possible if there are five charms in a bracelet?
A. 5
B. 6
C. 12
D. 24
- There are 10 people on each round table. A couple must sit next to each. How many possible arrangements are there?
A. $10!$
B. 20
C. 40, 320
D. 80, 640



Additional Activities

Solve the following problem.

- In how many ways can 7 people be seated at a round table if:
- they can sit anywhere?
 - three particular people must sit next to each other?
 - two people must not sit next to each other?



Lesson 3. Distinguishable Permutation



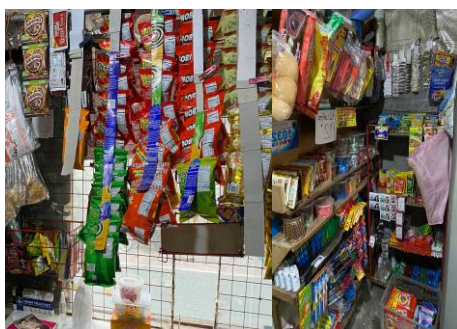
What's In

Let us have a quick recap of what we did last time!
Complete the table below:

	LINEAR PERMUTATION	CIRCULAR PERMUTATION
Definition		
Formula		
Sample in real-life		



What's New



Pictures taken from CANO STORE.

Observe the goods in a sari-sari store. Then, answer the following guide questions:

1. What are your experiences in buying goods in a sari-sari store?
2. What have you observed on the goods in a sari-sari store?
3. Are the goods in a sari-sari store illustrate permutation? Why or why not?



What is It

Not only in sari-sari stores, but this type of arrangement can also be seen in the cabinets of our kitchen. All the goods are arranged according to its similarities. This type of arrangement is known as distinguishable permutation in terms of Mathematics.



Definition: DISTINGUISHABLE PERMUTATION

It refers to the number of permutations of n objects of which n_1 are one of a kind, n_2 are of second kind, ..., and n_k are of a k th kind is given by:

Formula:
where $n = n_1 + n_2 + \dots + n_k$

In the simplest way, distinguishable permutation happens when similar objects are to be arranged.

Example 1: Suppose you are the owner of a sari-sari store and you want to put 12 pieces of canned goods in a row on the shelf. If there are 3 identical cans of meat loaf, 4 identical cans of tomato sauce, 2 identical cans of sardines, and 3 identical cans of corned beef, in how many different ways can you display these goods?

Explanation: The key word in the problem is 'identical' which means similar. Goods in a sari-sari store should be organized according to its similarities.

Illustration:

Type of Canned Goods	Symbol (n_k)	Number of canned goods
Meat loaf	n_1	3
Tomato sauce	n_2	4
Sardines	n_3	2
Corned beef	n_4	3
Total number of goods (n)		12

Solution:

$$\frac{n!}{n_1!n_2!\dots n_k!} = \frac{12!}{3!4!2!3!}$$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4!}{3!4!2!3!}$$

$$= \frac{\cancel{12} \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times \cancel{5} \times \cancel{4!}}{\cancel{3!} \cancel{4!} \cancel{2!} \cancel{3!}}$$

$$= 11 \times 10 \times 9 \times 8 \times 7 \times 5$$

$$= 277,200$$

Substitute $n = 12$ & the denominator

should be for the number of identical objects

Expand the numerator for simplification

Cancel similar values

Simplify

Answer: **There are 277,200 ways that you can display these goods.**



Example 2: How many distinguishable permutations can be formed from the letters of the word MARIKINA?

Explanation: On a first glance, it could be a linear permutation. Since letters can be repeated in a word, it is considered as distinguishable permutation.

Illustration:

LETTER IN THE WORD: 'MARIKINA'	Symbol (n_k)	NUMBER OF LETTERS
M	n_1	1
A	n_2	2
R	n_3	1
I	n_4	2
K	n_5	1
N	n_6	1
Total number of letters (n)		8

Solution:

$$\begin{aligned}
 \frac{n!}{n_1!n_2!\dots n_k!} &= \frac{8!}{1!2!1!2!1!1!} \\
 &= \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3!}{1!2!1!2!1!1!} \\
 &= \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times \cancel{2!}}{1!2!1!\cancel{2!}1!1!} \\
 &= \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3}{2} \\
 &= 10,080
 \end{aligned}$$

Substitute $n = 8$ & the denominator

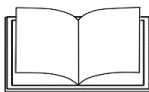
should be for the number of identical objects

Expand the numerator for simplification

Cancel similar values

Simplify

Answer: **There are 10,080 arrangements in the letters of the word Marikina.**



What's More

Answer the following problems.

- What is the number of possible arrangements of 9 books on a shelf where 4 are of the same kind, 3 are of the same kind, and 2 are of the same kind?
- There are 15 pigs in the pen which will undergo an experiment for the 3 different kinds of diet. Each diet should be used on 5 randomly selected pigs. In how many ways can the diets be assigned to the pigs?
- Determine the number of permutations of the following words:
 - GOOD
 - HEALTH
 - VITAMINS





What I Have Learned

Differentiate the three kinds of permutation by completing the details below.

	LINEAR PERMUTATION	CIRCULAR PERMUTATION	DISTINGUISHABLE PERMUTATION
Definition			
Formula			
Sample in real-life			



What I Can Do

“PERFORMANCE TASK: WHO AM I?”

Goal: Make a calligraphy of your name.

Role: You are a calligrapher.

Audience: Classmates and teacher.

Situation: Every person's name has a meaning behind it. You have high curiosity on your name's meaning.

Product:

- 1) Make a calligraphy of your name in a paper. Be more creative by making designs.
- 2) Research the meaning of your names or you may also talk to your parents/guardians the reason of having your name. Write it briefly under your calligraphy.
- 3) Answer with complete solution: Find the distinguishable permutation of your first name.

Standard for Success: Be guided with the scoring rubrics.

CRITERIA	3	2	1	0
Computation × 5	All representation and computations are correct and accurate.	Committed 1 error .	Committed 2-3 errors .	Committed 4 or more errors .



Creativity × 2	Artwork is exceptionally unique and interesting.	Artwork is generally unique and interesting.	Artwork is somewhat unique and interesting.	Artwork is not unique nor interesting.
Explanation × 3	Explained with in-depth research, logical and very clear.	Explained with basis of research and clear.	Incomplete details.	No output.



Assessment

Directions. Choose the letter that corresponds to the correct answer.

- Which of the following illustrates distinguishable permutation?
 - List of students
 - Choosing a partner for the party
 - Arrangement of books in the library
 - Checking of attendance in the record
- Which of the following is the formula for distinguishable permutation?
 - $\frac{n!}{(n-r)!}$
 - $\frac{n!}{(n-r)!}$
 - $\frac{(n-1)!}{2}$
 - $\frac{n!}{n_1!n_2!\dots n_k!}$
- In how many ways can 9 flags be arranged if 3 are of the same red, 4 are of the same blue and 2 are of the same yellow?
 - 1,260
 - 2,520
 - 15,120
 - 362,880
- Find the number of permutations of the telephone number 8700.
 - 48
 - 24
 - 16
 - 12
- How many distinguishable permutations can be made from the letters of the word *PEACE*?
 - 120
 - 60
 - 30
 - 24





Additional Activities

Determine the number of distinguishable permutations of your Learner's Reference Number.

Summative Assessment

Directions. Choose the letter that corresponds to the correct answer.

- It refers to the number of ways you arrange things where order is important.
A. event
B. combination
C. permutation
D. sample space
- Which of the following illustrates permutation?
A. Forming a committee from the members of the club
B. Checking of attendance in a class record
C. Selecting a player in basketball
D. Choosing a song to play
- How many ways can seven people be seated in a sofa if it can only accommodate five people?
A. 35
B. 1260
C. 2520
D. 5040
- How many ways can five different paintings be lined up in a wall?
A. 120
B. 60
C. 25
D. 5
- Evaluate: $P(9, 4)$
A. 13
B. 36
C. 324
D. 3024
- In a salvation bracelet, there are 7 different colored beads. How many arrangements of the beads are possible?
A. 360
B. 720
C. 2520
D. 5040
- There are 8 different books but only 5 can be occupied in your bag. How many arrangements can be made?
A. 6720
B. 3360
C. 120
D. 40



8. In how many ways can Aps and her five friends be seated in a round table?
- A. $3!$ C. $5!$
B. $4!$ D. $6!$
9. In how many ways can four cards from a deck of 52 cards be laid face up in a row?
- A. $P(4, 52)$ C. $P(52, 4)$
B. $P(52, -4)$ D. $P(52, 4)!$
10. If friends of 8 attend a party and three of them must sit together in a circular table, what must be the value of n ?
- A. 7 C. 5
B. 6 D. 4
11. Evaluate: $3! + 5!$
- A. 6 C. 126
B. 120 D. 40320
12. Chris needs to arrange 9 people in a circular table for her birthday's seat plan. How many arrangements can be made?
- A. 362,880 C. 20,160
B. 40,320 D. 20,106
13. How many ordered arrangements can be made from the letter in the word PROBABILITY?
- A. $\frac{11!}{2 \cdot 2}$ C. $\frac{11!}{2 \cdot 2!}$
B. $\frac{11!}{(2-2)!}$ D. $\frac{11!}{2!2!}$
14. You have 5 identical bottles and 4 identical towels to be given to your 9 friends. In how many ways can you distribute those 9 gifts to your 9 friends?
- A. 126 C. 15,120
B. 3024 D. 362,880
15. How many distinguishable permutations can be made from the letters of the word 'HEALTHY'?
- A. 5040 C. 2502
B. 2520 D. 2005



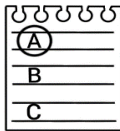
References

Coronel, Sr. Illuminada C., et.al. (2015) Growing Up with Math 10. FNB Educational, Inc. pp. 225-260

Nivera, Gladys C. PhD., et.al. (2015) Grade 10 Mathematics Patterns and Practicalities. Salesiana Books by Don Bosco Press, Inc. pp. 289-301

Rosario, Jose, et.al. (2018) Our World of Math Grade 10. Vibal Group, Inc. pp. 204-207.





Answer Key

What's More 1. 24; 2. 12; 3. 120; 4a. 5040; 4b. 720 What's I Have Learned 1. circular 2. (n - 1)! 3. two or 2 4. 6 Assessment 1. A; 2. C; 3. B; 4. C; 5. D Additional Activities a. 720; b. 144; c. 480 LESSON 3: DISTINGUISHABLE PERMUTATION What's In			
What's New 1. & 2. Answers may vary 3. Yes, because order matters What's More 1. 1260; 2. 756; 756; 3a. 12; 3b. 360; 3c. 20,160 What's I Have Learned			
What's I Know 1) C 2) C 3) A 4) D 5) A What's New Answer: 6720 ways 1. skin 2. lips 3. heart What's More A. 1. P; 2. NP; 3. NP; 4. P; 5. P B. 1. 720; 2. 5040; 3. 3; 4. 72; 5. undefined because n $\geq$ r C. 1. 5040; 2. 30,240; 3. 12 What's I Have Learned 1. Fact 2. Bluff 3. Fact 4. Bluff 5. Fact Assessment 1. B; 2. D; 3. D; 4. C; 5. A Additional Activities 1. 120; 2. 2520; 3. 15,120 LESSON 2: CIRCULAR PERMUTATION What's In 1. NP; 2. P; 3. P; 4. P; 5. NP What's New Answers may vary			

What's I Know 1) C 2) C 3) A 4) D 5) A What's New Answer: 6720 ways 1. skin 2. lips 3. heart What's More A. 1. P; 2. NP; 3. NP; 4. P; 5. P B. 1. 720; 2. 5040; 3. 3; 4. 72; 5. undefined because n $\geq$ r C. 1. 5040; 2. 30,240; 3. 12 What's I Have Learned 1. Fact 2. Bluff 3. Fact 4. Bluff 5. Fact Assessment 1. B; 2. D; 3. D; 4. C; 5. A Additional Activities 1. 120; 2. 2520; 3. 15,120 LESSON 2: CIRCULAR PERMUTATION What's In 1. NP; 2. P; 3. P; 4. P; 5. NP What's New Answers may vary			
Definition Arrangement of objects in linear form	Formula $(n - 1)!$	Sample in real-life Arrangement of objects in linear form	PERMUTATION
Definition Arrangement of objects in circular form	Formula $(n - 1)!$	Sample in real-life Arrangement of objects in circular form	CIRCULAR PERMUTATION





Development Team of the Module

Writer: Christlie Rose R. Nicolas (SRNHS)

Editors: Imelda B. Olvida, Principal (MHHS)
John Anthony P. Santos (SNNHS)
Melody F. Grimaldo (NHS)
Rosemarie P. Ortega (PHS)
Dona G. Aquino (KNHS)
Tiburcio A. Aplacador (FHS)
Wilfredo M. Junio (MNHS)

Internal Reviewer: Dominador J. Villafria (Education Program Supervisor-Mathematics)

Cover Illustrator: Joel J. Estudillo (SNNHS)

Layout Artist: Ferdinand B. Villegas (PHS)

Management Team:

Sheryll T. Gayola

Assistant Schools Division Superintendent
OIC, Office of the Schools Division Superintendent

Elisa O. Cerveza

Chief, CID
OIC, Office of the Assistant Schools Division Superintendent

Dominador J. Villafria

Education Program Supervisor-Mathematics

Ivy Coney A. Gamatero

Education Program Supervisor– LRMS

For inquiries or feedback, please write or call:

Schools Division Office- Marikina City

Email Address: sdo.marikina@deped.gov.ph

191 Shoe Ave., Sta. Elena, Marikina City, 1800, Philippines

Telefax: (02) 682-2472 / 682-3989



City of Good Character
DISCIPLINE • GOOD TASTE • EXCELLENCE